

MSET006**Question 1.**

[Tg:@abusat_760] In the xy -plane, the graph of $y = 4 - (x + 4)^2$ is the image of the graph of $y = 2 - x^2$ after which of the following transformations?

- A) A translation of 2 units up and 4 units to the right
- B) A translation of 2 units down and 4 units to the right
- C) A translation of 2 units up and 4 units to the left
- D) A translation of 4 units down and 2 units to the right

Question 2.

$$9 - (x + 1)^2 = 0$$

[Tg:@abusat_760] If $x = \mu$ is a solution to the equation above, which of the following must be true?

- A) $\mu < -5$
- B) $-5 < \mu < 3$
- C) $3 < \mu < 4$
- D) $4 < \mu$

Question 3.

[Tg:@abusat_760] In the xy -plane, the point with coordinates $(a, 2)$ and $(b, 2)$ lie on the graph of the equation $y = 5x^2 - 3$. If $a \neq b$, what is the value of $b^2 - a^2$?

- A) -3
- B) 0
- C) 5
- D) 17

Question 4.

$$x^2 + 3x - 1 = 0$$

[Tg:@abusat_760] Which of the following is a solution to the equation above?

- A) $\frac{-3+\sqrt{13}}{2}$

B) $\frac{3-\sqrt{13}}{2}$

C) $\frac{-3+\sqrt{7}}{2}$

D) $\frac{3-\sqrt{7}}{2}$

Question 5.

$$f(x) = (x - a)(x - b)$$

[Tg:@abusat_760] The function f is defined above, where a and b are integer constants. If $f(1) > 0$, $f(4) < 0$, and $f(7) > 0$, what is one possible value of $a + b$?

Question 6.

[Tg:@abusat_760] If $g(x) = 30 + 2(x + 1) + x^2$, what is the value of $g(5)$?

A) 65

B) 66

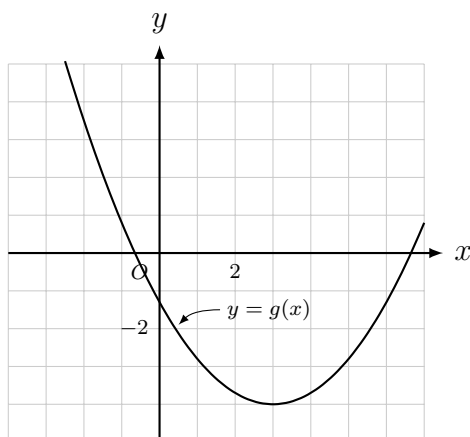
C) 67

D) 77

Question 7.

[Tg:@abusat_760] The function $h(t) = -16t^2 + 32t + 6$ can be used to model the height of a ball, in feet, t seconds after it is thrown into the air. The function can be rewritten in the form $h(t) = -a(t - h)^2 + k$, where k is the maximum height of the ball, in feet. What is the value of k ?

Question 8.



[Tg:@abusat_760] The function g is shown in the xy -plane above, and the function f (not shown) is defined by $f(x) = x^2 + 4x + 3$. Which of the following statements must be true?

- I. The minimum value of f is greater than the minimum value of g .
 - II. The y -intercept of f is less than the y -intercept of g .
 - III. The functions f and g intersect on the x -axis.
- A) I only
- B) II only
- C) III only
- D) I and II only

Question 9.

$$h(t) = -19.2t^2 + 9.8t + 7$$

[Tg:@abusat_760] At a baseball home run hitting contest, the trajectory of one home run was examined. The baseball's height h , in feet above the ground, t seconds after being hit is modeled by the quadratic function above. Which of the following is the best interpretation of the number 7 in this context?

- A) The number of seconds it took the baseball to reach its maximum height.
- B) The number of seconds it took the baseball to hit the ground.
- C) The number of feet above the ground the baseball was 19.2 seconds after it was hit.
- D) The number of feet above the ground the baseball was when it was hit.

Question 10.

$$x^2 - x = 0$$

[Tg:@abusat_760] Which of the following are the solutions to the equation above?

- A) $x = \frac{1}{4} \pm \sqrt{\frac{1}{2}}$
- B) $x = \frac{1}{4} \pm \sqrt{\frac{9}{4}}$
- C) $x = \frac{1}{2} \pm \sqrt{\frac{1}{4}}$
- D) $x = \frac{1}{2} \pm \sqrt{\frac{9}{4}}$

Question 11.

$$\frac{3}{2x-1} - \frac{x}{5} = 0$$

[Tg:@abusat_760] In the equation above, $2x - 1$ could equal which of the following?

- A) -6
- B) -5
- C) 4
- D) 8

Question 12.

[Tg:@abusat_760] The function f is defined by $f(x) = -ax^2 + bx + 7$ where a and b are positive constants. Which of the following points CANNOT be on the graph of $y = f(x)$ in the xy -plane?

- A) $(-2, 4)$
- B) $(0, 9)$
- C) $(1, 5)$
- D) $(1, 6)$

Question 13.

[Tg:@abusat_760] The height h , in feet, of a toy rocket above the ground t seconds after it is launched can be modeled by the function $h(t) = -9.8t^2 + 24.5t$. Which of the following methods can be used to find m , the maximum height, in feet, of the rocket above the ground?

- A) Rewrite h as $h(t) = -9.8(t - 1.25)^2 + 15\frac{5}{16}$. The number $15\frac{5}{16}$ represents m .
- B) Rewrite h as $h(t) = -t(9.8t + 24.5)$. The number 24.5 represents m .
- C) Rewrite h as $h(t) = -9.8t(t - 2.5)$. The number 2.5 represents m .
- D) Compute $h(0) = -9.8(0)^2 + 24.5(0)$ to find $h(0) = m$.

Question 14.

$$3x^2 - 21x + 36 = 0$$

[Tg:@abusat_760] If $x = k$ represents a solution to the quadratic equation above, what is one possible value of k ?

Question 15.

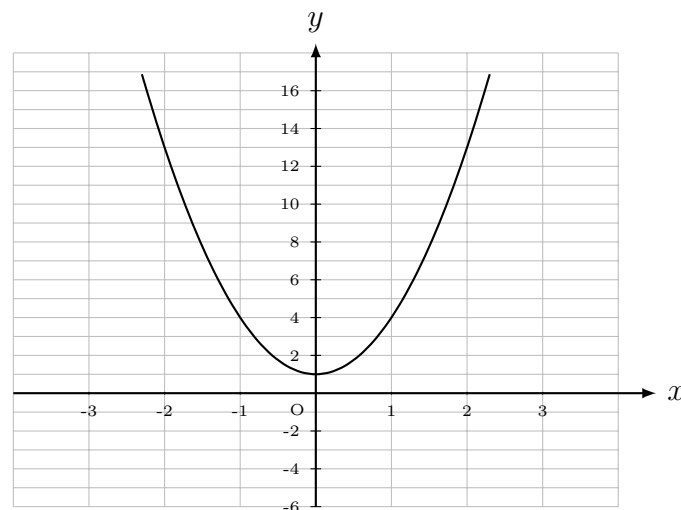
$$(x + 4)^2 = 18x$$

[Tg:@abusat_760] Which of the following is a value of x that satisfies the equation above?

- A) -4
- B) -2
- C) 8
- D) 18

Question 16.

[Tg:@abusat_760] The graph of $f(x) = ax^2 + b$, where a and b are integer constants, is shown below.



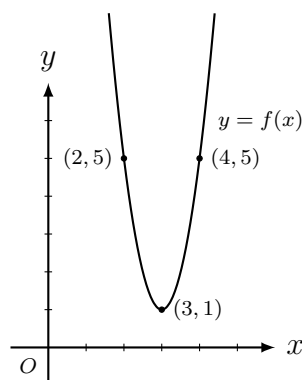
What is the value of b ?

Question 17.

$$\frac{x}{x-3} = \frac{2x}{2}$$

[Tg:@abusat_760] Which of the following represents all the possible values of x that satisfy the equation above?

- A) 0 and 2
- B) 0 and 4
- C) -4 and 4
- D) 4

Question 18.

[Tg:@abusat_760] The graph of the function f in the xy -plane above is a parabola. Which of the following defines f ?

- A) $f(x) = 4(x - 3)^2 + 1$
- B) $f(x) = 4(x + 3)^2 + 1$
- C) $f(x) = (x - 3)^2 + 1$
- D) $f(x) = 3(x + 3)^2 + 1$

Question 19.

[Tg:@abusat_760] If $f(x) = 5x^2 - 3$ and $f(x + a) = 5x^2 + 30x + 42$, what is the value of a ?

- A) -30
- B) -3
- C) 3
- D) 30

Question 20.

[Tg:@abusat_760] In the xy -plane, the graph of the polynomial function f crosses the x -axis at exactly two points, $(a, 0)$ and $(b, 0)$, where a and b are both positive. Which of the following could define f ?

- A) $f(x) = (x - a)(x - b)$
- B) $f(x) = (x + a)(x + b)$
- C) $f(x) = (x - a)(x + b)$
- D) $f(x) = x(x - a)(x - b)$

Question 21.

[Tg:@abusat_760] If $y = 3x^2 + 6x + 2$ is graphed in the xy -plane, which of the following characteristics of the graph is displayed as a constant or coefficient in the equation?

- A) y -coordinate of the vertex
- B) x -intercept(s)
- C) y -intercept
- D) x -intercept of the line of symmetry

Question 22.

$$h(x) = -16x^2 + 100x + 10$$

[Tg:@abusat_760] The quadratic function above models the height above the ground h in feet, of a projectile x seconds after it had been launched vertically. If $y = h(x)$ is graphed in the xy -plane, which of the following represents the real-life meaning of the positive x -intercept of the graph?

- A) The initial height of the projectile
- B) The maximum height of the projectile
- C) The time at which the projectile reaches its maximum height
- D) The time at which the projectile hits the ground

Question 23.

[Tg:@abusat_760] Which of the following intervals contains all solutions of $x^2 \leq 2x + 35$?

- A) $[2, 5]$
- B) $[-5, 7]$
- C) $[0, 5]$
- D) $[7, 0]$

Question 24.

[Tg:@abusat_760] In the xy -plane, which of the following is an equation whose graph is the graph of $y = 2(x - 2)^2 + 2$ translated five units horizontally to the right?

- A) $y = 7(x + 3)^2 + 2$
- B) $y = 2(x + 3)^2 + 7$
- C) $y = 2(x - 7)^2 + 7$
- D) $y = 2(x - 7)^2 + 2$

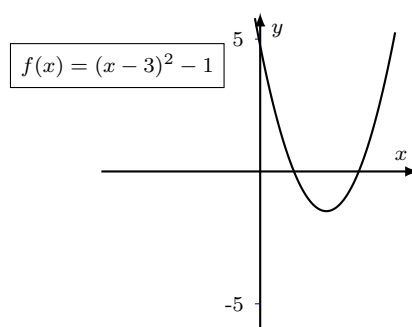
Question 25.

[Tg:@abusat_760] Alex throws rocks vertically with a velocity of 96 feet per second. When t seconds, the rock height is h feet from the ground. The relation between h and t is about,

$$h = 96t - 16t^2$$

Which interval is the rock height at least 80 feet?

- A) $1 \leq t \leq 2$
- B) $1 \leq t \leq 5$
- C) $2 \leq t \leq 3$
- D) $3 \leq t \leq 6$

Question 26.

[Tg:@abusat_760] Which of the following is an equivalent form of the equation of the graph shown in the xy -plane above, from which the coordinates of x can be identified as constants in the equation?

- A) $y = (x - 1)(x - 2)$
- B) $y = (x - 2)(x - 3)$
- C) $y = (x - 4)(x - 2)$
- D) $y = (x + 2)(x + 4)$

Question 27.

$$f(x) = \sqrt{(x-2)^2 + 2x - 3}$$

[Tg:@abusat_760] For what is the least value of the range of Function f ?

Question 28.

$$f(x) = cx^2 + 30$$

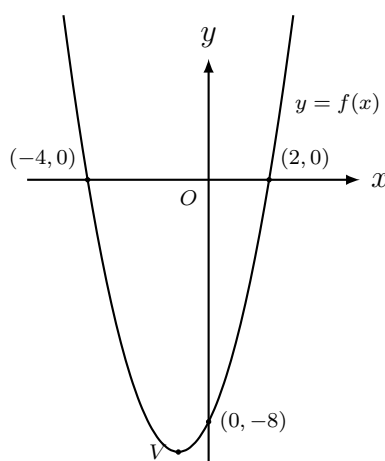
[Tg:@abusat_760] For the function f defined above, c is a constant and $f(3) = 12$. What is the value of $f(-3)$?

- A) -12
- B) -2
- C) 0
- D) 12

Question 29.

[Tg:@abusat_760] If $(ax+3)(bx+5) = 35x^2 + kx + 15$ for all values of x , and $a+b = 12$, what are the two possible values for k ?

- A) 46 and 50
- B) 15 and 35
- C) 21 and 25
- D) 5 and 7

Question 30.

[Tg:@abusat_760] The figure above shows the graph in the xy -plane of the function $f(x) = x^2 + 2x - 8$. Which of the following is an equivalent form of the function f that includes the coordinates of vertex V as constants in the function?

- A) $f(x) = (x + 1)^2 - 9$
- B) $f(x) = x(x - 2) - 8$
- C) $f(x) = x(x - 2)(x + 4)$
- D) $f(x) = x(x + 2)(x - 4)$

Question 31.

[Tg:@abusat_760] For what value of x is the expression

$$\frac{2}{(x-6)^2 + 4(x-7) + 8}$$

undefined?

Question 32.

[Tg:@abusat_760] The graph in the xy -plane of the function g has the property that y is always greater than or equal to -3 . Which of the following could be g ?

- A) $g(x) = -x^2 - 3$
- B) $g(x) = (x - 3)^2$
- C) $g(x) = |x| - 3$
- D) $g(x) = (x - 3)^3$

Question 33.

$$3k^2 - 18k + 12 = 0$$

[Tg:@abusat_760] What is the product of all values of k that satisfy the equation above?

- A) 3
- B) 4
- C) $3\sqrt{5}$
- D) $6\sqrt{5}$

Question 34.

[Tg:@abusat_760] In the equation $3(x-5)^2 + 7 = ax^2 + bx + c$, a , b and c are constants. If the equation is true for all values of x , what is the value of c ?

Question 35.

$$y = x^2 - 12x + 35$$

[Tg:@abusat_760] The equation above is a quadratic equation. Which of the following equivalent forms of the equation displays the x -intercepts of the parabola in the xy -plane as constants or coefficients?

- A) $y + 1 = (x - 6)^2$

- B) $y - 35 = x^2 - 12x$
C) $y = (x - 5)(x - 7)$
D) $y = x(x - 12) + 35$

Question 36.

$$g(x) = 2x^2 - kx + 14$$

[Tg:@abusat_760] In the xy -plane, the graph of the function above contains the point $(4, -2)$. What is the value of k ?

Question 37.

$$y = k(x - 4)(x + 12)$$

[Tg:@abusat_760] In the quadratic equation above, k is a constant such that $k \neq 0$. When the equation is graphed in the xy -plane, the graph is a parabola with vertex (m, n) . Which of the following is equal to n ?

- A) $-24k$
B) $-36k$
C) $-48k$
D) $-64k$

Question 38.

[Tg:@abusat_760] In the xy -plane, a parabola defined by the equation $y = (x - 8)^2$ intersects the line defined by the equation $y = 36$ at two points, P and Q . What is the length of PQ ?

Question 39.

[Tg:@abusat_760] What are the solutions to $5x^2 + 30x + 15 = 0$?

- A) $x = -2 \pm 2\sqrt{6}$
B) $x = -2 \pm \sqrt{6}$
C) $x = -3 \pm \frac{\sqrt{160}}{10}$
D) $x = -3 \pm \sqrt{6}$

Question 40.

[Tg:@abusat_760] If the expression $(6x^2 - 7x + 5) - 3(x^2 - 5x + 4)$ is written in the form $ax^2 + bx + c$, what is the value of b ?

Question 41.

[Tg:@abusat_760] If $x^3 - 4x^2 + 3x - 12 = 0$, what real value is a solution for x ?

Question 42.

[Tg:@abusat_760] In the quadratic equation $x^2 - 3t = \frac{v}{3}$, t and v are constants. What are the solutions for x ?

- A) $x = \frac{v}{3} \pm \frac{\sqrt{v^2 + 4t}}{3}$
- B) $x = \frac{v}{3} \pm \frac{\sqrt{v^2 + 36t}}{6}$
- C) $x = \frac{v}{6} \pm \frac{\sqrt{v^2 + 108t}}{6}$
- D) $x = \frac{v}{6} \pm \frac{\sqrt{v^2 + 4t}}{6}$

Question 43.

$$5x^2 + 7x - 6 = 0$$

[Tg:@abusat_760] If a and b are two solutions to the equation above and $a < b$, which of the following is the value of $b - a$?

- A) $\frac{3}{5}$
- B) $\frac{6}{5}$
- C) $\frac{8}{5}$
- D) $\frac{13}{5}$

Question 44.

$$d = 8t^2 + vt + h$$

[Tg:@abusat_760] The equation above gives the distance, d , in meters, a projectile is above the ground t seconds after it is released with an initial velocity of v meters per second from an initial height of h meters. Which of the following gives v in terms of d , t , and h ?

- A) $v = \frac{d - h}{t} - 8t$
- B) $v = \frac{d + h}{t} - 8t$
- C) $v = \frac{d - h + 8}{t}$
- D) $v = d + h - 8t$

Question 45.

$$v = \frac{h - m}{t} + 4.9t$$

[Tg:@abusat_760] For an object thrown straight upward, the formula above gives the relationship between v , the initial speed in meters per second, t the time in seconds after the object was thrown, h , the height after t seconds, and m , the initial height from which the object was thrown. Which of the following expresses h , in terms of v , t and m ?

- A) $h = -4.9t^2 + vt - m$
- B) $h = -4.9t^2 + vt + m$
- C) $h = -4.9t^2 - vt + m$
- D) $h = 4.9t^2 - vt - m$

Question 46.

[Tg:@abusat_760] Let the function f be defined by $f(x) = (x - 1)(x + 7)$. Which of the following is an equivalent form of f in which the minimum value of f appears as either a coefficient or a constant?

- A) $f(x) = x^2 - 7$
- B) $f(x) = x^2 + 6x - 7$
- C) $f(x) = (x + 3)^2 - 16$
- D) $f(x) = (x - 3)^2 - 20$

Question 47.

$$y = a(x - 3)(x + 5)$$

[Tg:@abusat_760] In the quadratic equation above, a is a nonzero constant. The graph of the equation in the xy -plane is a parabola with vertex (s, t) . Which of the following is equal to t ?

- A) $-2a$
- B) $-10a$
- C) $-15a$
- D) $-16a$

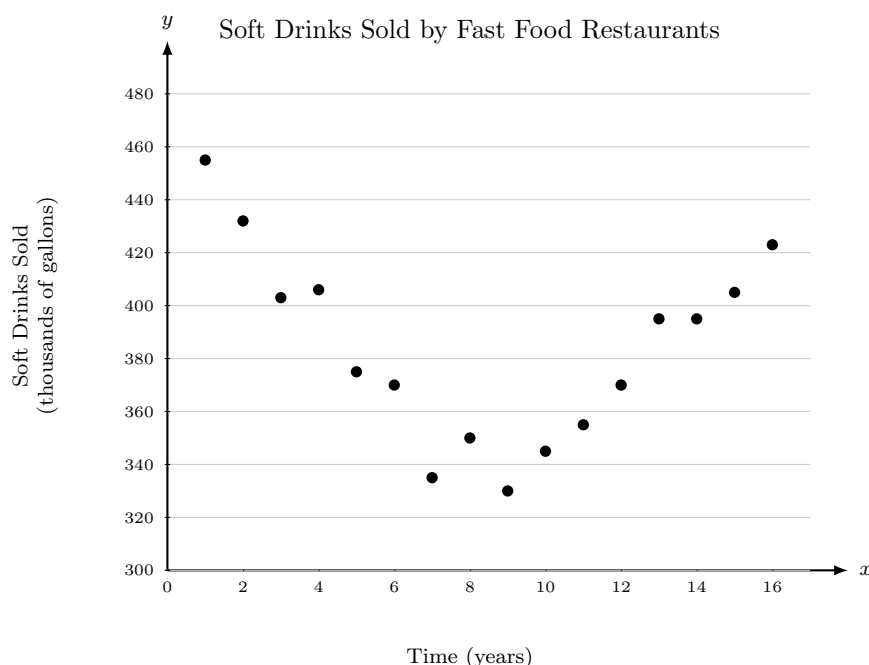
Question 48.

$$a^3(a^2 - 25) = -144a$$

[Tg:@abusat_760] If $a > 0$, what is one possible solution to the equation above?

Question 49.

[Tg:@abusat_760] The scatterplot below shows the number of soft drinks sold, in thousands of gallons, by fast food restaurants over a 15-year period.



Of the following equations, which best models the data in the scatterplot?

- A) $y = 2.138x^2 - 38.44x + 483.27$
- B) $y = 2.138x^2 + 38.44x - 483.27$

C) $y = -2.138x^2 - 38.44x - 483.27$

D) $y = -2.138x^2 - 38.44x + 483.27$

Question 50.

[Tg:@abusat_760] What are the solutions of the quadratic equation $-3x^2 - 3x + 18 = 0$?

A) $x = -3$ and $x = 2$

B) $x = -3$ and $x = -2$

C) $x = 3$ and $x = -2$

D) $x = 3$ and $x = 2$

Answer Key

1	2	3	4	5	6	7	8	9	10
C	B	B	A	7,8,9	C	22	D	D	C

11	12	13	14	15	16	17	18	19	20
A	B	A	3 or 4	C	1	B	A	C	A

21	22	23	24	25	26	27	28	29	30
C	D	B	D	B	C	0	D	A	A

31	32	33	34	35	36	37	38	39	40
4	C	B	82	C	12	D	C	D	8

41	42	43	44	45	46	47	48	49	50
4	C	D	A	B	C	D	3 or 4	A	A